# Do Joint Language-Audio Embeddings Encode Perceptual Timbre Semantics?

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## Overview

• Joint language-audio models are widely used for retrieval, captioning, and text-guided audio generation.
• Encode source source or event（nouns)
• Unclear whether these models encode **perceptual timbre semantics** (e.g., bright, rough).
• We evaluate four embedding models using both **human-rated instrument timbre** and **DSP-controlled timbre manipulations.**

## Background

• Timbre descriptors (adjectives such as bright, dark) are central in music production, effects control, and sound design.
• Joint language-audio embeddings map audio and text into a shared space.
• Excel at identifying sound sources/events.
• Their representation of **subtle perceptual sound qualities,** especially timbre, has not been systematically studied.

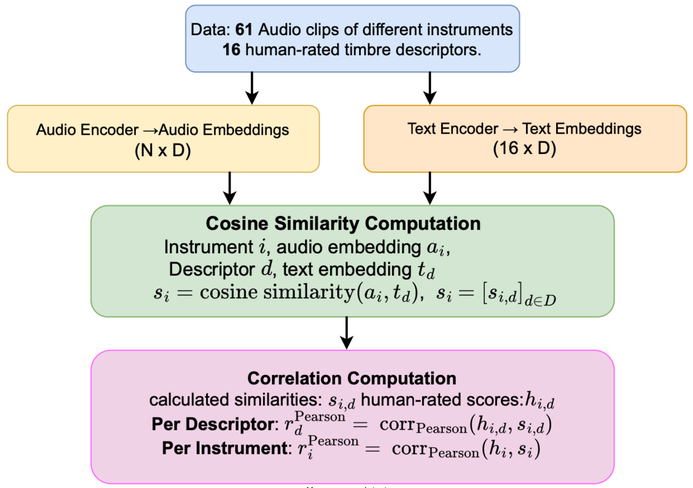
## Research Questions

• **Do joint language-audio embeddings reflect human-perceived timbre semantics?**
• How well do different models align with:
- **Human timbre ratings** across diverse instruments?
- **Controlled timbre changes** produced by EQ and reverb?
• Which model provides the most perceptually grounded timbre representation?

## Experiment 1: Instrumental Timbre Semantics

**Dataset**: Jiang et al.'s CCMusic-Database-Instrument-Timbre dataset (37 Chinese, 24 Western instruments, 16 descriptors rated by trained listeners).

| Instrument | slim | bright | dim | sharp | thick | thin |
|---|---|---|---|---|---|---|
| violin | 5.2 | 5.3 | 3.4 | 4.1 | 4.1 | 3.7 |
| viola | 3.5 | 4.4 | 4.4 | 3.4 | 6.2 | 2.9 |




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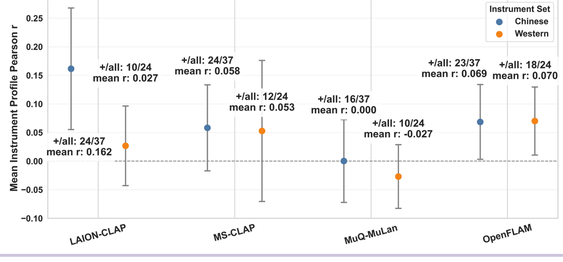
**Descriptor-Level Correlation Across Models**

| Descriptor | LAION-CLAP | MS-CLAP | MuQ-MuLan | OpenFLAM |
|---|---|---|---|---|
| bright | 0.22 | -0.15 | 0.40 | 0.41 |
| clear | 0.35 | -0.21 | -0.27 | 0.27 |
| coarse | -0.25 | 0.22 | 0.08 | -0.08 |
| dim | 0.11 | 0.31 | -0.13 | -0.17 |
| dry | 0.15 | -0.09 | 0.04 | -0.01 |
| harmonious | 0.10 | 0.11 | 0.10 | -0.19 |
| hoarse | 0.18 | 0.26 | 0.17 | -0.06 |
| plump | 0.16 | -0.09 | -0.14 | -0.20 |
| pure | 0.09 | -0.04 | -0.11 | 0.04 |
| sharp | 0.30 | -0.16 | -0.15 | 0.32 |
| slim | -0.03 | -0.12 | 0.41 | 0.27 |
| soft | 0.21 | -0.13 | -0.01 | 0.04 |
| thick | -0.19 | 0.25 | -0.14 | -0.23 |
| thin | -0.15 | -0.29 | 0.12 | 0.26 |
| turbid | 0.27 | 0.38 | -0.07 | -0.04 |
| vigorous | 0.35 | 0.19 | -0.48 | -0.29 |
| Mean correlation | 0.117 | 0.027 | -0.011 | 0.021 |
| # +correlation | 12/16 | 7/16 | 7/16 | 7/16 |

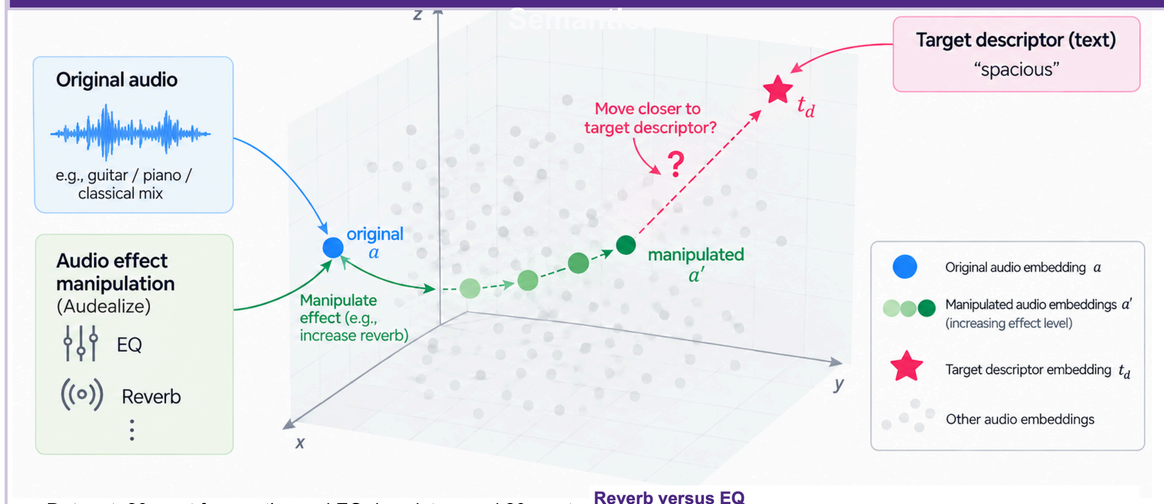
Pearson r: 0.4, 0.2, 0.0, −0.2, −0.4

**Instrument-Level Performance by Model and Culture**

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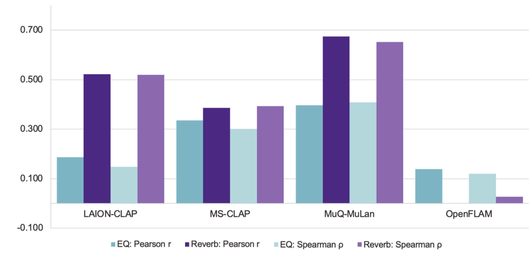



## Experiment 2: Audio Effects Timbre

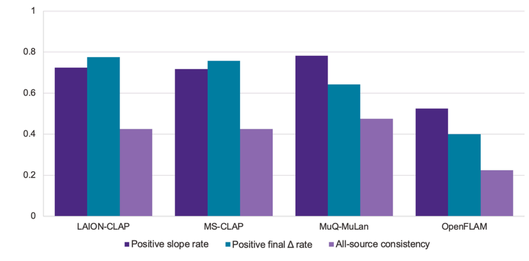



• **Dataset**: 20 most frequently used EQ descriptors and 20 most frequently used reverb descriptors.
• **Effect** parameters: descriptor-specific settings from Audealize.
• **Stimuli**: guitar, piano, and classical ensemble.
• **Manipulations**: original audio plus EQ or reverb at five intensity levels: 0.2, 0.4, 0.6, 0.8, and 1.0.
• Measured the change in audio-text similarity relative to the original audio.
• **Evaluation**: trajectory slope and final similarity change; consistency across source clips; Pearson correlation (linearity); Spearman correlation (monotonicity).

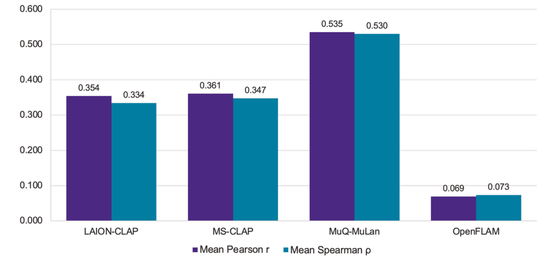
**Reverb versus EQ**




**Model-level trends**




**Trajectory correlations**




## Conclusion

• Overall alignment with human timbre ratings is weak.
• Most models align more strongly with reverb than EQ.
• Some descriptors align with selected models, but no model captures timbre semantics consistently.

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